Stage 1: is the board alive?

You add: the ESP32-C3 Super Mini and a USB-C cable. Nothing else is connected.

You learn: how to upload a sketch and read the Serial Monitor, and that a pin can be an output or an input.

Steps

1. Set up the Arduino IDE as described in the main README.
2. Plug the board into the computer.
3. Open 01_blink.ino, choose the board and the port, and press Upload.
4. Open Tools > Serial Monitor at 115200 baud.
5. Press and hold the BOOT button on the board for a second or two.

What you should see

The blue light on the board blinks, on for half a second and off for half a second. The Serial Monitor shows:

```
===== Stage 1: board test =====
Chip          ESP32-C3 (simulated), revision 4
Processor     160 MHz
Flash memory  4 MB
Light on GPIO8, BOOT button on GPIO9
The light should now blink. Press and hold BOOT to see the last column change.

tick    1   light ON    BOOT button not pressed
tick    2   light off   BOOT button not pressed
tick    3   light ON    BOOT button not pressed
tick    4   light off   BOOT button PRESSED
tick    5   light ON    BOOT button PRESSED
tick    6   light off   BOOT button not pressed
```

This output, like every example in this guide, comes from running the sketch on a computer in the simulator. On your board the chip line reads a little differently.

The stage is passed when the light blinks, the lines keep coming, and the last column changes while you hold BOOT.

What is going on

- **Upload.** The Arduino IDE turns the sketch into a program for the chip and writes it into the board's flash memory through the USB cable. The program then starts by itself every time the board gets power.
- `setup()` **runs once,** `loop()` **runs for ever.** Every Arduino sketch has these two.
- **The Serial Monitor** shows the text the board sends back over the same USB cable. It is the main way to see what a sketch is doing.
- **Output.** `pinMode(LED_PIN, OUTPUT)` lets the sketch drive GPIO8. The light is soldered between 3.3 volts and that pin, so it lights when the pin is LOW. That is why the sketch has `LED_ON = LOW`.
- **Input.** `pinMode(BUTTON_PIN, INPUT_PULLUP)` lets the sketch read GPIO9. A small resistor inside the chip pulls the pin HIGH. The button joins it to ground, so pressed reads LOW.

Try this

- Change `delay(500)` to `delay(100)` and upload. The light blinks five times as fast.
- Change `LED_ON` to `HIGH`. The words ON and off in the monitor no longer match the light.

If it does not work

See Uploading and the Serial Monitor.

Next: stage 2.